

Lab: Phenomenological Analysis of Predictive Processing

Learning Goal: Apply phenomenological methods to analyze predictive processing claims.

Part 1: Phenomenological Mapping

Exercise: For each phenomenological concept, explain the Active Inference formalization and assess the mapping:

Phenomenological Concept	Active Inference Formalization	Quality of Mapping (1-5)	Why?
Intentionality (Husserl)	Generative model predictions		
Ready-to-hand (Heidegger)	Low prediction error		
Body schema (Merleau-Ponty)	Proprioceptive generative model		
Lifeworld (Husserl)	Prior beliefs (D vector)		
Intersubjectivity (Husserl)	Multi-agent inference		

Write your response here

Part 2: First-Person Prediction Error Report

Learning Goal: Use phenomenological description to analyze your own prediction errors.

Exercise: Over one day, note three instances where prediction errors became phenomenologically salient:

1. Describe the situation in phenomenological terms (what did you experience?)
2. Analyze it in Active Inference terms (what model was violated? What was the prediction error?)
3. How did you resolve it? (model update, action, precision change?)

Write your response here

Part 3: The Realism Debate

Learning Goal: Evaluate the direct/indirect realism debate through predictive processing.

Exercise: Consider a simple perceptual experience — seeing a red apple on a table.

1. Describe this experience from a **direct realist** perspective (you simply see the apple).
2. Describe it from an **indirect realist** perspective (you perceive an internal representation of the apple).
3. Describe it from the **Active Inference** perspective (your generative model predicts apple-observations; the prediction is confirmed by low prediction error).
4. Which account best captures the phenomenology — the "what it's like" — of seeing the apple?

Write your response here

Part 4: The Illusion Challenge

Learning Goal: Analyze perceptual illusions through the FEP lens.

Exercise: Choose a well-known perceptual illusion (e.g., Müller-Lyer, rubber hand, hollow mask):

1. Describe the illusion phenomenologically
2. Explain why the generative model produces the illusory percept
3. If perception is "controlled hallucination," what distinguishes this illusion from normal perception?
4. What does this tell us about the relationship between perception and reality?

Write your response here

Part 5: Reflection

In 250 words, reflect: Does the convergence between phenomenology and predictive processing represent a genuine theoretical unification, or a superficial similarity between two very different projects (first-person description vs. third-person explanation)?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Conceptual mapping	Phenomenology ↔ Active Inference
2	First-person analysis	Prediction errors in experience
3	Philosophical argument	Realism debate
4	Applied analysis	Illusions and model failure
5	Meta-philosophical reflection	Unity of approaches